Task 1 Report: BestBuy.ca Review Scraping and Sentiment Analysis

Product analysed: Samsung Essential F320 27" 1080p HD IPS LED Monitor (LS27F320GANXZA), BestBuy.ca SKU 19277986. Reviews dated up to October 2026.

1. Objective and approach

The goal was to collect public customer reviews for a BestBuy Canada product with at least 50 reviews, apply sentiment analysis, and turn the results into recommendations for stakeholders.

The work followed five steps: collect public reviews, clean and structure the data, score sentiment, tag the aspects customers talk about, and summarise the drivers of satisfaction and dissatisfaction. All analysis was done in Python (pandas, BeautifulSoup, VADER, matplotlib) in the submitted notebook `task1_scrape_sentiment.ipynb`.

2. Data collection

Product choice. The Samsung F320 monitor was chosen because it has more than 50 public reviews. Two earlier candidates were rejected because they had too few reviews (the HomePod mini 2 had none, and a Samsung Odyssey G9 had 13).

Fields captured. Primary key, title, review text, date (YYYY-MM-DD), rating (out of 5), source, and reviewer display name. Extra fields added: overall sentiment, aspect tags, sentiment score, which filters surfaced the review, and a flag for promotional reviews.

Result. 55 unique reviews after de-duplication (110 review cards were parsed, because every review appears twice on the saved page). Every review has a rating and a date. 42 reviews contain written text and 13 are rating-only (a star rating with no body text). The rating-only reviews are kept because the rating is valid data. Each review gets a stable primary key (a hash of title, text, reviewer and date), so a review that appears under several filters is stored once.

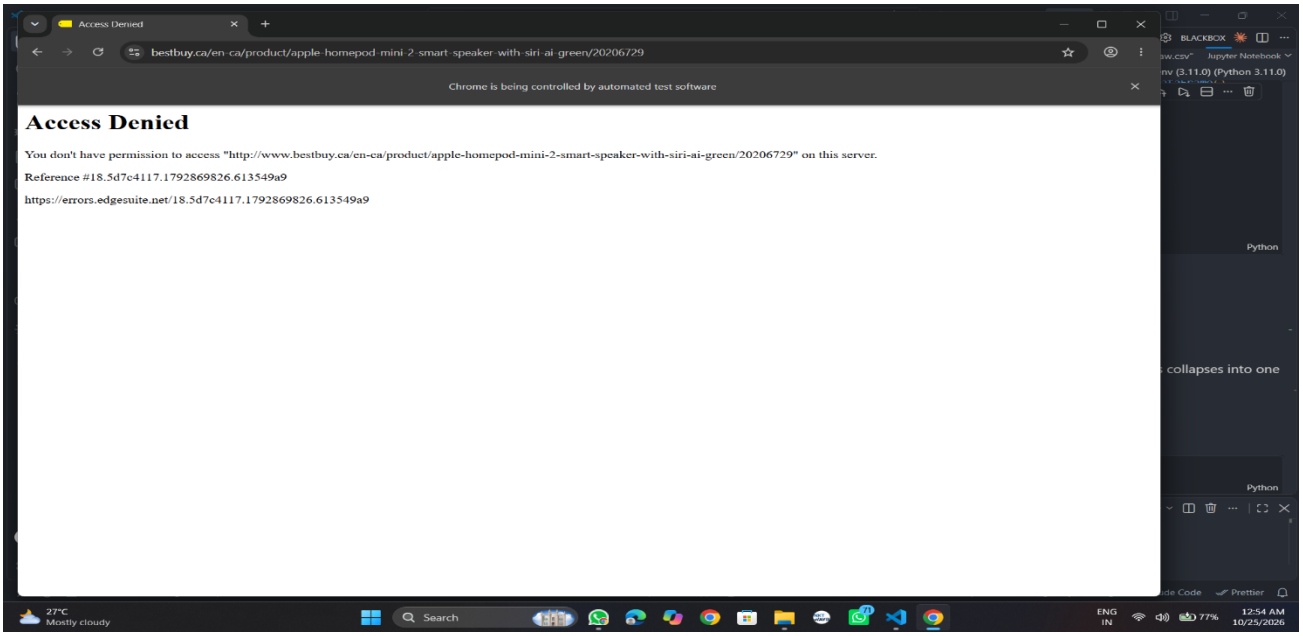
Filters and pagination. The notebook contains Selenium logic that applies the five sort filters (Most Relevant, Most Helpful, Newest, Highest Rating, Lowest Rating) and clicks "Show More" until no new reviews load. See section 3 for what happened when it ran against the live site.

Compliance. Only publicly visible review content was collected. No login was used, the robots.txt file was checked, and traffic was throttled with random delays. Reviewer names are public display names only.

3. Scraping challenges and solutions

What happened. An automated Chrome session (Selenium) was blocked by the site's bot protection and returned an "Access Denied" page for the product URL. The block was not circumvented, because that would conflict with the site's policy and the assignment's instruction not to violate it. Instead, the public review pages were opened in a normal browser session, all reviews were loaded with "Show more", the pages were saved, and the saved HTML was parsed with BeautifulSoup. The Selenium and JSON-endpoint code remains in the notebook as the intended automated method.

Challenge	How it shows up	Solution
Bot detection / access denied	HTTP 403 or an "Access Denied" page	Respect the block. Collect via normal browsing, or use an official API or a licensed data feed for scale
IP blocking and rate limiting	403/429 responses, blank pages	Random 2 to 5 second delays, low request volume, exponential back-off retries. At scale: rotating proxies where the site terms allow it
CAPTCHA	"Verify you are human" page	Slow the pace and let a person solve it in a visible browser. For production, prefer an official data partnership
Dynamic JavaScript content	Static HTML contains no reviews	Selenium waits and "Show More" clicks, or call the same JSON endpoint the page uses
Layout changes	Selectors return nothing	Ordered fallback selectors, regex-based rating extraction, a selector-inspector helper, and alerts when parsed rows drop
Transient failures	Timeouts, random exceptions	`@retry` decorator with back-off, per-view error handling, and a raw-data checkpoint file
Legal and ethical	Terms of use, privacy	robots.txt check, public data only, no personal data beyond a display name



Access Denied returned to the automated browser

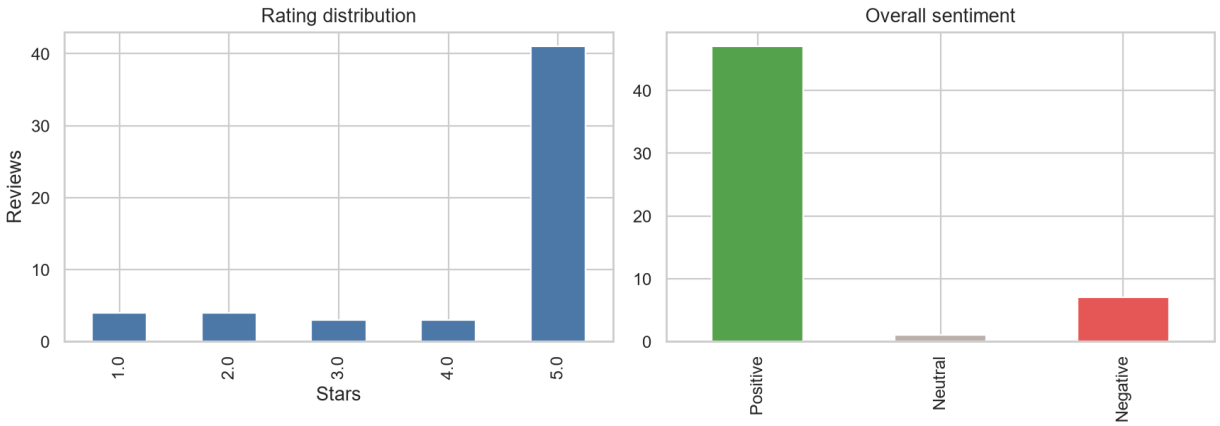
4. Sentiment method

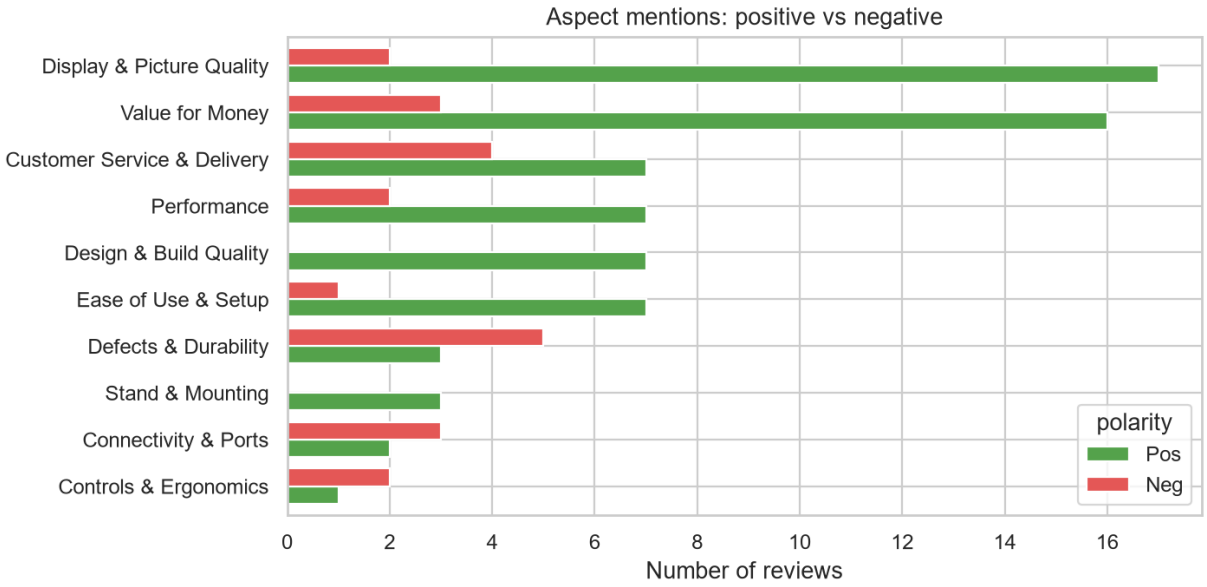
1. **Pre-processing:** URLs and symbols removed, whitespace normalised. The promotional tag that BestBuy adds to some reviews was detected, recorded in a separate column, and removed from the text so it does not influence sentiment. Reviews without body text are scored from the star rating and the title only.
2. **Overall sentiment:** the VADER score (-1 to +1) is averaged with the star rating, rescaled to the same range. Labels: Negative below -0.15, Neutral up to 0.15, Positive above.
3. **Aspect tags:** each review is split into clauses, and every clause that mentions an aspect keyword gets a polarity. This produces tags such as `Display & Picture Quality (Pos)`. The keyword list covers display, build quality, stand, controls, setup, ports, performance, value, defects and service.

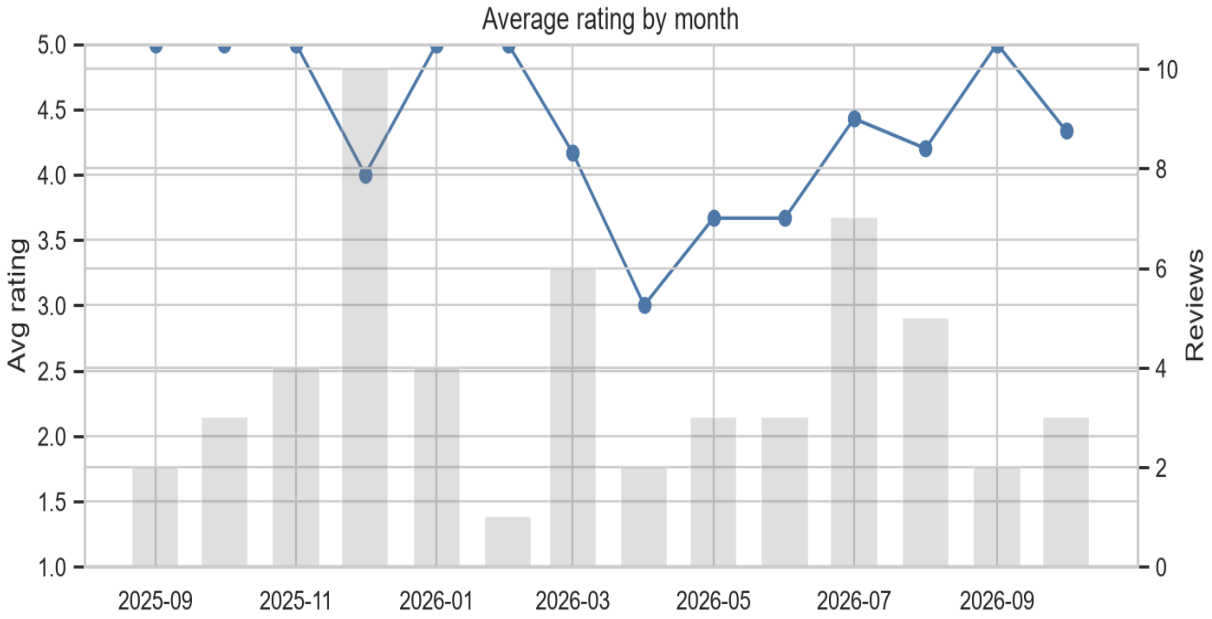
Why the rating is blended in. Among the 42 reviews with written text, text-only VADER agreed with the star rating on 83% of them (35 of 42). The disagreements came mostly from mixed or very short reviews, where the wording is neutral or positive even though the rating is low, and where short factual praise such as "Great deal" carries little emotional wording. Blending in the rating corrects these cases. Note that the blended result agrees with the rating by construction, so the 83% figure is the honest measure of text-only accuracy.

5. Results

- **Reviews analysed:** 55 | **Average rating:** 4.33 out of 5. Of the 55 reviews, 42 contain written text and 13 are rating-only, and the aspect analysis uses the 42.
- **Rating distribution:** 5 stars: 41, 4 stars: 3, 3 stars: 3, 2 stars: 4, 1 star: 4.
- **Overall sentiment:** 85% Positive (47 reviews), 13% Negative (7 reviews), 2% Neutral (1 review).
- **Promotional vs organic:** 7 reviews carried the promotion tag, with an average rating of 4.71, against 4.27 for the 48 organic reviews. The sample is small, so this is an observation and not a conclusion.




Aspect mentions: positive vs negative
Display & Picture Quality
Value for Money
Customer Service & Delivery
Performance
Design & Build Quality
Ease of Use & Setup
Defects & Durability
Stand & Mounting
Connectivity & Ports
Controls & Ergonomics
polarity
Pos
Neg
0
2
4
6
8
10
12
14
16
Number of reviews


Average rating by month
Avg rating
Reviews
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
10
8
6
4
2
0
2025-09
2025-11
2026-01
2026-03
2026-05
2026-07
2026-09



6. Insights for stakeholders

Top drivers of satisfaction

- **Display and picture quality:** 17 positive mentions (89% of mentions about it). Customers call the picture clear and bright, and this is the product's strongest selling point.
- **Value for money:** 16 positive mentions (84%). Price is a frequent reason for purchase.
- **Customer service and delivery:** 7 positive mentions (64%), although the same aspect also has 4 negative mentions, so it is mixed.

Top drivers of dissatisfaction

- **Defects and durability:** 5 negative mentions (62% of mentions). Examples: a dead area in the top right of the screen, missing pixels, large white spots on arrival, and a unit that would not power up consistently.
- **Customer service and delivery:** 4 negative mentions (36%). Example: the wrong order was given at pickup, followed by a wait of several days.
- **Connectivity and ports:** 3 negative mentions (60%). Example: a monitor advertised with two HDMI ports where only one worked.

Additional finding from reading the 2 and 3-star reviews. Two reviews criticise the power button, which sits on the back of the monitor and also toggles the on-screen menu. Both still rated the product 3 stars because the picture quality was good, so the text-based scoring did not rank this issue highly. It is a design problem that a numeric ranking understates.

7. Recommendations

1. **Reduce panel defects.** Add quality checks for dead pixels, dead zones and brightness uniformity before shipping, and review the return process for screen faults.
2. **Fix the controls.** Move the power button to a reachable position and separate it from the menu toggle.
3. **Verify and state ports.** Test both HDMI ports in final quality control and state the exact port versions on the product page.
4. **Improve order fulfilment.** Address order accuracy at pickup and shorten the wait for corrected orders. Check packaging for transit damage.
5. **Lead with strengths in marketing.** Picture quality and price are the strongest positives, so feature them on the product page.
6. **Treat promotional reviews separately** when reporting a headline rating, because they rate somewhat higher.

8. Limitations

- **Small sample:** 55 reviews, of which only 7 are Negative. The dissatisfaction drivers rest on a handful of reviews, so they are indicative and not statistically firm.
- **Skewed ratings:** 75% of reviews are 5 stars, which leaves little negative text to learn from. In addition, 13 of the 55 reviews (24%) carry a star rating but no written text, so only 42 reviews feed the aspect analysis.
- **Possible mixed products:** a few reviews mention a TV, streaming features, or buffering, which suggests that reviews may be shared across Samsung variants. These should be checked manually before drawing product-specific conclusions.
- **Lexicon sentiment:** VADER misses sarcasm and short factual praise, and aspect detection is keyword-based, so some tags are wrong (for example, a clause about a previous monitor can be tagged as a defect of this one). Tag precision on a hand-checked sample of 20 reviews: 28 of the 33 tags that could be checked against the visible text were correct (85%). The errors came from negation (a missing feature described with the words does not have was tagged positive), from a keyword that carried no opinion about its aspect (a review of an HDMI disconnection tagged under value for money), and from the word can't being read as negative in a positive review. Some obvious aspects were also missed (for example a complaint about mounting holes, or an HDMI problem), so recall is lower than precision; it was not measured. Two of the 20 reviews were cut off in the printed sample, so only tags with visible supporting text were counted.
- **Collection method:** pages were saved in a normal browser session after automated access was blocked, so collection is manual and not repeatable at scale.

9. Next steps

- Use an official API or a licensed review feed for larger, repeatable collection.
- Fine-tune a transformer sentiment model, or use a zero-shot or LLM aspect extractor, validated on a hand-labelled sample.
- Collect reviews for several competing monitors to benchmark aspects such as picture quality and defect rates.
- Schedule periodic collection to track whether defect complaints fall after fixes.